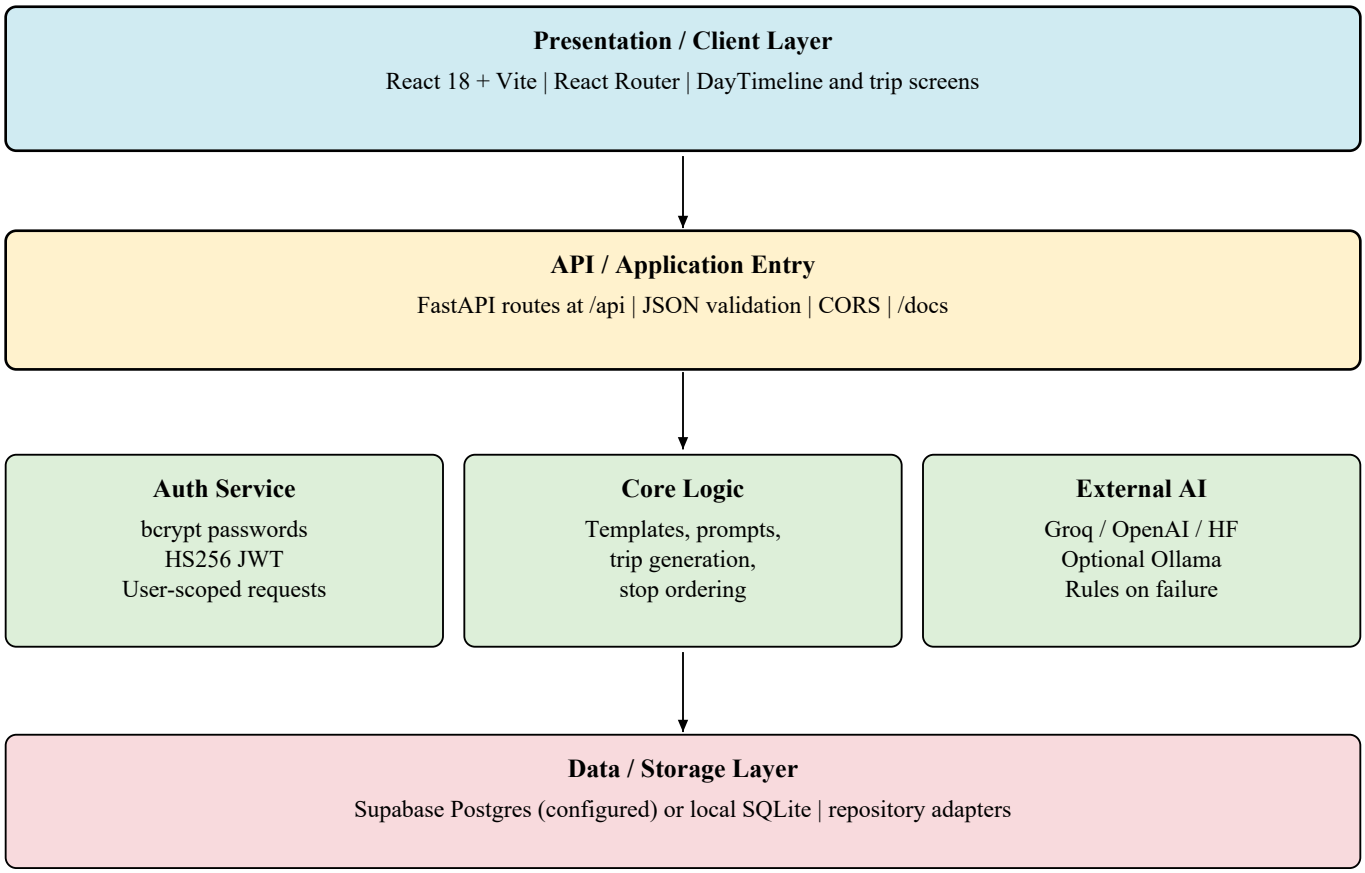
# Solution Architecture

| | |
|---|---|
| **Date** | 23 October 2026 |
| **Team ID** | 0827CY231029 |
| **Project Name** | Waypoint - AI Travel Itinerary Planner |
| **Maximum Marks** | 5 Marks |

## Solution Architecture Diagram

A React single-page application calls one modular FastAPI backend. The application uses repository adapters for Supabase or SQLite; the internal services are modules within this backend.




The entry box represents the FastAPI API service. A separate gateway or load balancer is not configured in the source project.

**Solution Architecture (continued)**

## Component Description Table

| Component Name | Description / Role in Architecture | Technologies Used |
|---|---|---|
| Presentation Layer | Collect inputs, display curated trips and daily timelines, manage saved trips and show public plans. | React 18, Vite, React Router, Framer Motion, Lucide, CSS |
| API / Application Entry | Expose /api routes; validate inputs; configure CORS and error responses; publish /docs. | FastAPI, Pydantic, Uvicorn/Gunicorn |
| Auth Service | Hash passwords, verify credentials, issue signed seven-day JWTs and require tokens on private routes. | bcrypt; python-jose; HTTPBearer |
| Core Logic | Prepare prompts, generate day plans, rank destinations, order stops and format exports. | Python service modules; curated catalog; trip routers |
| AI Integrations | Select a configured model provider; receive JSON output; use local rules on failure. | Groq, OpenAI, Hugging Face or optional Ollama; httpx |
| Database | Persist users, preferences, trips, days, activities and recommendation logs. | Supabase Postgres or SQLite; repository adapters |

## Configuration and Availability

Database mode is selected at startup from Supabase configuration. SQLite is a local mode, not a live failover mirror of production data. AI generation can continue using local rules when the selected model is unavailable.

Project sources: backend main.py, store.py, config.py, security.py, services/ and database schemas; frontend App.jsx.